Chapter 1

Supplementary Information for Chapter 2: Helium Step Heat Experiment on Goethite in Hydrothermal Conditions

1.1 Supplementary Verification: The Naïve/Informed Bracketing Method for the Encapsulated, Irradiated Win-06-01b Step Heat Experiment

This supplement documents the naïve/informed bracketing method: an independent, more bespoke treatment of the HT-Win-06-01b dataset that explicitly tracks the helium content of the ampoule step by step, rather than bounding it as in the main text's "treatment as uncertainty" approach (§**??**, §**??**). It is retained here solely as a secondary, independent verification of the main text's results and conclusions, not as the primary basis for them: the results of the two methods' distilled datasets and derived quantities agree, corroborating the main text's simpler treatment. This supplement covers only the naïve/informed bracketing dataset and the bespoke, step-by-step corrections applied to it (the Raw Data and Corrections Columns through Totals sections below); the isothermal-permeation-fit methodology and cooling-tube numerical model that establish the ampoule's elevated permeability – the evidentiary basis for the main text's conservative literature-permeability assumption, and a prerequisite for the corrections applied below – are described in full in the main text (§**??**).

1.1.1 Summary

Helium isotope measurements of evolved gasses from hydrothermal steps heats of glass-encapsulated irradiated HT-Win-06-01b goethite of the Flinders Ranges (**monteiro2017**), were acquired at Caltech on a Thermo Fisher Helix-MC-Plus^{10K} mass spectrometer in peak hopping mode from January 31st, 2024 to March 13th, 2024. Raw data from these measurements are found on the first tab "Raw Data and Corrections" of the "IrrWin0601bTube012624FullDataBJT.xlsx" file. Because the sample was encapsulated in an imperfectly helium-transmissive silica glass ampoule, experimental steps at temperatures at or above 170°C were bracketed by low temperature (<120°C) steps where the helium in goethite at high pH2O was effectively immobile for experimental timescales. For 3He,

the resulting low-temperature measurements were above blank for the majority of the experiment and were entirely controlled by the process of permeation through silica glass, and therefore were used to estimate the amount of residual helium within the ampoule between high-temperature experimental steps. These corrections are carried out in the "Raw Data and Corrections" tab using permeability values derived internally within the experiment from the "Naïve Apparent Lambdas" tab. Diffusivities calculated using the equations of **fechtigkalbitzer1966** for 3He and 4He can be found on the "Diffusivities" tab. During the step heat experiment, 116 measurements of high 3He MM gas standard were analyzed to track instrumental drift of 3He and 4He sensitivities and to quantify internal uncertainties using reproducibility. For the final step, the glass ampoule was broken and the remainder of the helium was degassed in a resistance furnace at 1100°C. Data from this final step can be found on the "Totals" tab of "IrrWin0601bTube012624FullDataBJT1.xlsx".

Raw Data and Corrections Columns

Raw Data

"Nominal (Heated) Time" refers to the time programmed for the step heat. These times are used for both calculating apparent diffusivities with the equations of **fechtigkalbitzer1966** and for permeation lag corrections.

The "Watlow Time" column gives the last timestamp recorded in the Watlow temperature log at the target temperature. It is always slightly higher than the nominal time because the recorded time includes the initial time required to heat the sample up to the programmed temperature: typically around 30 seconds to two minutes. This column was generated by a python script that read these values from the Watlow text file temperature records. It was used in early stages of data analysis to identify when Watlow errors within Labview had held the sample for longer than the programmed step duration. When these discrepancies were identified, the "Nominal Time" was updated to reflect the time the sample was held at the programmed temperature. These times are not directly used for any of the data reduction.

The "Total Time" column is the difference between the analysis timestamp and the previous analysis timestamp. These timestamps are recorded during the inlet of evolved, purified Helium into the Helix-MC-Plus^{10K}, which immediately precedes the pump out of the lamp cell chamber to the turbo pump. Therefore, the "Total Time" represents the total amount of time that helium was allowed to permeate through the silica glass ampoule into the UHV lamp cell, including both the time the sample spent "at temperature" and the time that the sample was cooling to or at room temperature (24-26°C). Since the pump out times were only 1 minute and always occurred when the sample was coolest (and permeation was slowest), permeation from the ampoule during pump out is neglected in the data reduction. Less than one thousandth of the helium remaining in the ampoule would permeate out within one minute at room temperature (Eq. (**??**), Chapter **??**, with log permeability (ln(K)) at 26°C of 23.88 ln(atoms/cm*atm*s)). Due to the formatting of the time stamps, these values have a 1-minute resolution.

The "Cooling Time" is the difference between the "Total Time" and "Nominal Time". It is greater than or equal to the time to purify evolved gas in preparation for analysis: ~26 minutes. This value is used for permeation corrections when low-temperature permeation steps with cooling times greater than 26 minutes are used.

Abundances, 4He Blank Corrections, Measured 4He/3He

Sensitivity and Uncertainty Calculations

3He and 4He sensitivities were calculated for each analysis of the MM standard gas, assuming that each standard aliquot contained 157.50±0.41(2σ) nccSTP 4He and 3.567±0.040(2σ) pccSTP 3He. These values were obtained from sensitivities from a known amount of a standard gas with similar helium abundances and with the signals from 10 aliquots of the new MM standard tank in June 2023.

1 sigma uncertainties related to internal reproducibility were estimated for each sensitivity calculation as the rolling standard deviation of sensitivities calculated from 5 MM standards. In other words, the internal reproducibility of the sensitivity was estimated as the standard deviation of each calculated sensitivity and the 2 sensitivities immediately before and after. Uncertainties related to external reproducibility from the amount of 3He and 4He in each standard aliquot were added to the total uncertainty as well.

Sensitivities and their uncertainties were interpolated to times between standard analyses using a Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) from the SciPy package in python. These interpolated sensitivities and uncertainties were used to convert the 3He and 4He signals (cps and fA respectively) to absolute atomic abundances and uncertainties in units of pccSTP and nccSTP respectively. The sensitivities and interpolations are graphically displayed in the "STDs" tab in a plot generated using the Matplotlib package in python.

Helium-4 Blank Corrections

Throughout the experiment, low temperature permeation steps yielded unpredictable 4He signals that did not exponentially decay like the 3He signals did. This effect is most obvious when considering analyses where cooling times were constant (~1560 seconds). In these cases, 3He signals always decreased exponentially whereas the 4He signals often increased (or decreased) erratically. A blank was measured at the beginning of the experiment and showed negligible blank for 3He (regressing to below 0 cps at the inlet time) and a very low value for 4He ($4.84\text{x}10^{-2}$ fA). Low temperature permeation steps sometimes yielded much higher signals despite failing to conform to the expectation of exponential decay that is observed in the same steps' 3He measurements. For instance, the 4He signal in a cool permeation step at 105°C from reference file He30509.txt was $7.93\text{x}10^{0}$ fA, even when the high temperature experimental step preceding it (He30502.txt) yielded a similar value: $8.28\text{x}10^{0}$ fA. For the same pair of analyses, the 3He signal matched expectations and varied by two orders of magnitude. 4He from permeation steps never fit the equation for cumulative permeation, $m_{cum} = n_0 \left(1 - e^{-\lambda t}\right)$ (Eq. (**??**), Chapter **??**), and thus could not be used to infer the permeability for 4He or for the residual amount of 4He remaining in the ampoule (n_0). For comparison, the same equation always fit the low temperature permeation steps well for 3He when the cooling time of each analysis was constant.

The erratic behavior in the 4He data is best explained by a shifting blank during the experiment, which was not measured or quantified directly. It was largest in the middle of the experiment and returned to low values by the end of the experiment that were similar to the blank measured before the experiment. Because the blank for 4He was not well controlled and appeared to have affected many of the values observed even in the high temperature experimental steps, a blank correction was deemed necessary. The 4He data of the permeation steps are interpreted here to be dominated by blanks, so high temperature experimental steps were coarsely blank corrected by subtracting the average 4He signal of the subsequent permeation steps. The entirety of this correction was conservatively added to the 2σ uncertainty of the experimental data for 4He. One blank corrected

measurement was within error of zero after permeation corrections and was not plotted. The permeation lag was estimated for the 4He data using the 3He data and assuming that the 4He permeability was 0.91357±0.14603 (2σ) that of the 3He permeability. These relative permeability values for 3He and 4He were measured at Caltech for a similar silica glass ampoule half-filled with water and equilibrated with ~ 0.1 torr of tank helium. This process is described in greater detail in the Permeation Lag Corrections section.

Some 3He permeation steps that followed especially low-yield experimental steps were within error of baseline 3He cps values (similarly to the measured blank at the beginning of the experiment). These low measurements always preceded by measurements where 3He was quantified above blank, so permeation lag corrections were calculated using those values instead. In these cases, uncertainty from the permeation lag corrections dominate the uncertainty of the diffusivities, but do not affect the interpretation of the dataset.

Bracketing Method Ratios

4He/3He ratios are presented with reference to the irradiated Winsor Goethite bulk ratio (~1000 or ~1000 Ra) and as standalone values. Uncertainties are propagated in quadrature from the measured 3He and 4He abundances and from the bulk ratio, when applicable. The blank and permeation corrected ratios have the same general shape as the uncorrected ratios. The ratios plateau around a cumulative 3He fraction of 0.06 at a bulk normalized 4He/3He ratio of 0.75. The early plateau in 4He/3He ratios with 3He fraction occurs similarly in the vacuum step heat experiments of the same sample, in agreement with previous studies (**monteiro2017**; **farley2024**), suggesting a relatively low loss of radiogenic helium for the Winsor Goethite. The vacuum and hydrothermal experiments differ in the bulk-normalized 4He/3He ratio at the plateau, with the hydrothermal experiment plateauing at a lower 4He/3He ratio of 0.75 times the bulk ratio, whereas the vacuum experiments plateau at the bulk ratio. This could be explained by a 4He-retentive inclusion in the hydrothermal experiment of this work similar to those discussed by **vasconcelos2013**, or may reflect trapping of 3He in newly formed hematite during the early steps of the vacuum experiment.

Permeation Lag Corrections for 3He

Helium evolved from the irradiated goethite must first permeate through the silica glass ampoule into the high vacuum of the lamp cell to be analyzed, a process which is non-instantaneous. Measurements can therefore underestimate the helium production of the sample during a step, as some of the produced helium will remain within the ampoule after a high temperature step, and measurements can also overestimate the amount of produced helium if there is some amount of helium remaining from previous step heats within the ampoule at the beginning of a new experimental step. To address this effect, we quantify the amount of helium remaining within the ampoule between high temperature experimental steps, and apply corrections using the rearranged form of Eq. (**??**) (Chapter **??**), solved for m:

$$m = n_0 + \omega - n_t$$

Where ω is the true helium production of the sample in a high temperature experimental step, n_0 is the amount of helium within the ampoule at the beginning of the step, n_t is the amount of helium within the ampoule at the end of the experiment, and m is the amount measured. Values of ω and m are most often within 10% of one another but differ by up to 47.2% for 3He within this dataset. n_0 and n_t are tracked in the "3He n_0 (pccSTP)" column, where n_t for any measurement is the n_0 value of the next row.

At low temperatures (<120°C), when no new helium is evolved from the sample, the measured value is simply the difference between the helium contents of the ampoule before the measurement and after (Eq. (**??**) with $\omega = 0$):

$$m = n_0 - n_t$$

In this case, all the helium measured is a result of helium permeation through the silica glass ampoule. Helium permeation is a first order process that follows Eq. (**??**) (Chapter **??**), with $\lambda = KGRT$:

$$n_t = n_0 e^{-KGRTt}$$

Where K is the temperature-dependent permeability coefficient, G is a geometric factor which can be determined through measurements of the ampoule's inner and outer diameter, R is the gas constant, T is the temperature in Kelvin, and t is the time in seconds. For a cylindrical ampoule, the value of G can be calculated with measurements of the inner and outer radii using Eq. (**??**) (Chapter **??**). Inner and outer radii were measured using the amscope program at 11x zoom setting. These radii were both measured on opposing sides of the ampoule where there were air bubbles, allowing for the accurate sighting of the inner walls of ampoule. The inner diameter, *(2a)* was measured at 505 and 540 microns; the outer diameter *(2b)* was measured at 787 and 832 microns (Fig. **??**). The geometric factor G was calculated for both pairs of measurements and averaged to give a value of G of 6708.6 cm^{-1}.

For permeation steps, Eqs. (**??**) (with $\omega = 0$) and (**??**) apply because the temperature is too low to evolve any helium from the sample, and the $\omega = 0$ form of Eq. (**??**) can be substituted into Eq. (**??**) to relate a measurement to n_0. This equation is calculated in the column entitled "3He n0 calc (pccSTP) (m/(1-eˆ-λt))". Errors in the measured value m and the permeability K are propagated in quadrature. This is the informed-bracketing relation of Eq. (**??**) (Chapter **??**), expressed with Eq. (**??**)'s exponential form for ${}^{He}F$:

$$n_0 = \frac{m}{(1-e^{-KGRT})}$$

The "3He n_0 (pccSTP)" column tracks the accepted value of n_0 (and $n_{t,;}$ which is n_0 in the subsequent column) for ultimate use in permeation corrections using Eq. (**??**). Permeation steps colored in pink and magenta are "Informed Bracket" measurements. They set the accepted value of n_0 at the time of their measurement using Eq. (**??**). In other words, the "3He n_0 (pccSTP)" column inherits the value calculated by Eq. (**??**) in the "3He n0 calc (pccSTP) (m/(1-eˆ-λt))" column for permeation steps colored pink, whereas the permeation steps colored darker magenta set the value of n_0 with an altered version of Eq. (**??**) because they cooled to room temperature and sat permeating for longer than 1560 seconds before measurement. Their respective n0 values are calculated using an altered form of Eq. (**??**) (Eq. (1.1)) in a column entitled "3He n0 calc long cool":

$$n_0 = \frac{m}{\left(1 - (e^{-K_{hot}GRT_{hot}t_{hot}} * e^{-K_{cold}GRT_{cold}t_{cold}}\right))} \tag{1.1}$$

T_{cold} is 26°C, K_{cold} is measured internally in a process described in more detail below, and t_{cold} is equal to the value in the "Cooling time" column minus 1560 seconds, because the effective permeabilities for the warmer temperatures used here were calculated from measurements that cooled for 1560 seconds.

Most experimental steps within this study are separated by multiple permeation steps that each decrease the value of n_0. Values in the "3He n_0 (pccSTP)" column are color coded according to how they are calculated. Uncolored cells denote the default method of tracking the decreasing values of n0, which is handled with Eq. (**??**). The calculated n_0 value for uncolored step i in the ith row is $n_0^i = n_0^{i-1} e^{-RG(KTt)_{i-1}}$. This default behavior is followed when cooling times for the previous step were 26 minutes, to avoid applying a value for the effective permeation constant 'K' that doesn't capture the additional effects of long cooling. It is advantageous to use this approach because it allows for the n0 value to be set by earlier measurements with larger signals, avoiding potential errors related to near-blank measurements. Errors associated with K and n_0^{i-1} are propagated through Eq. (**??**) in quadrature.

Values in the "3He n_0 (pccSTP)" that are colored brown or tan are propagated by adding or subtracting measured amounts of helium from the previous or subsequent step's determined value of "3He n_0 (pccSTP)", and simply adding the uncertainty from the previous or subsequent n_0 and the corresponding measured value of helium m. Tan cells are calculated by adding that cell's measured value m to the subsequent row's value of n_0, while the brown cells are calculated by subtracting the previous row's measured value m from its value of n_0.

Permeation steps/series in rows colored grey at the end of the experiment were arbitrarily long and were considered to effectively empty the tube of any helium, so that the value of n_0 for the subsequent cell was 0.

Permeation steps that are colored blue were taken at room temperature and were used to calculate the tube's permeability at room temperature *(K_{cold})* for use in Eq. (1.1).

Calculating the Permeability of the Ampoule at Room Temperature (K_{cold})

Three measurements were taken of the experimental ampoule permeating at room temperature; corresponding to the blue rows. The determined values can be found after the final columns of the "Raw Data and Corrections" tab. These measurements solve for the permeation coefficient K at 26°C using Eq. (**??**) solved for K, given by Eq. (1.2).

$$K = -\frac{\ln(n_t) - \ln(n_0)}{GRTt} \tag{1.2}$$

Where n_t is the value of n_0 in the "3He n_0 (pccSTP)" column of the next row. Uncertainties for n_0 and n_t are propagated in quadrature to K and are relatively large, however the values of K for the three measurements are tightly clustered. The mean value of K determined from the three measurements is taken as K_{cold} and the standard deviation of the three measurements is taken as the 1σ uncertainty. These steps all added/subtracted the measured values m to determine the value of n_0 so that there was no dependence K_{cold} on their n_0 values. Their n_0 values are thus color-coded tan or brown.

The first of these three measurements (H30413) was slightly dependent on the value of K_{cold}, as all the permeation steps in that series had cooled for longer than 26 minutes and required Eq. (1.1) to determine n_0. This series was treated subtly different than the rest of the permeation steps, since it both required and contributed a value of K_{cold}. The informed bracketing step He30418 is given bold yellow text in addition to being colored dark magenta to signify that the n_0 was determined using Eq. (1.1). For this step, K_{cold} and its uncertainty was taken to be the average and standard deviations of the other two determinations of K_{cold} ($2.68\text{x}10^{10} \pm 8.50\text{x}10^{9}$). The rest of the magenta "informed bracketing" steps used the average and standard deviation of all three measured values of K_{cold} ($2.56\text{x}10^{10} \pm 7.63\text{x}10^{9}$). All measured values of K_{cold} are within uncertainty of one another.

The permeation coefficient K_{cold} determined here is *not* an effective permeation coefficient, because there was no change in temperature at the end of the measurement. K_{cold} is plotted along with the effective permeation coefficients determined with the naïve bracketing method in a smaller plot, but it was not incorporated into the regression and error of the estimation of the effective permeabilities used in Eq. (**??**) for the informed bracketing method, only for the darker magenta informed bracketing steps using Eq. (1.1).

Permeation Lag Corrections for 4He

Because of blank-level 4He signals for the permeation steps, 4He abundances within the ampoule could not be independently estimated. Instead, they were estimated using the 3He n_0 abundances in the "3He n_0 (pccSTP)" column immediately following high-temperature experimental steps, assuming a high-uncertainty but temperature independent ratio of permeabilities $\left(\frac{K_{3He}}{K_4He}\right)$. This approach utilizes the steady-state n_t value n_{ss} already established in Chapter **??**: for any long high-temperature experimental step where the helium source is assumed constant and the permeation coefficient is known, Eq. (**??**) gives $n^{ss} \leq m/(\lambda t - 1)$, which for these long steps ($\lambda t \gg 1$) is well approximated by $n^{ss} \approx \omega/(\lambda t)$, the exact steady-state balance between the sample's helium production rate Ω and permeative loss ($n^{ss} = \Omega/\lambda$, with $\omega = \Omega t$ the total production over the step).

Taking the ratio of the steady state abundances for 3He and 4He within the ampoule:

$$\left.\frac{n_t^{4He}}{n_t^{3He}}\right|_{ss} = \frac{\omega^{4He}\lambda^{3He}}{\omega^{3He}\lambda^{4He}}$$

Consider that $\lambda = KGRT$ and G, R, and T are constants, the above expression reduces. Additionally, the value for ω is not known, but it is related to the measured value in an experimental step according to Eq. (**??**): $\omega = m + n_t - n_0$. For long experimental steps at high temperature, $\omega \gg n_t,\ n_0$, so $\omega \approx m$. Furthermore, even when n_t and/or n_0 are significant relative to the measured value m, their relative significances $(\frac{n_t}{m}, \frac{n_0}{m})$ would be expected to be very similar for both isotopes, because their permeabilities through silica glass are similar and both isotopes experienced the same heating schedule that would affect their abundances similarly. Therefore, both isotopes would be expected to differ from the assumption that $\omega \approx m$ in a similar fashion. In other words, the assumption $\frac{\omega^{4He}}{\omega^{3He}} \approx \frac{m^{4He}}{m^3He}$ is a much stronger assumption than $\omega \approx m$. Making this assumption and reducing:

$$\left.\frac{n_t^{4He}}{n_t^{3He}}\right|_{ss} \approx \frac{m^{4He}K^{3He}}{m^{3He}K^{4He}}$$

Which can be solved for n_t^{4He}

$$n_t^{4He} \approx \frac{m^{4He}K^{3He}}{m^{3He}K^{4He}} \bullet n_t^{3He} \tag{1.3}$$

In this work, $\frac{K_{3He}}{K_4He} = 1.10 \pm 0.19$ (2σ) and is not treated as a function of temperature, as conservatively measured by (**thyer2026**). When used in any equation in this work, its uncertainty is always propagated through quadrature.

Eq. (1.3) is solved in the column entitled "4He n0 by ratio (nccSTP)". Since Eq. (1.3) only applies to the n_t value of a high temperature experimental step (which is the same as the n_0 value of the permeation step immediately following), it is only used to assign n_0^{4He} values for the permeation step immediately following each high-temperature experimental step. $\frac{m^{4He}}{m^{3He}}$ refers to

the ratio of measured helium values of the high temperature step, and the n_t^{3He} term is n_0^{3He} of the permeation step immediately following a high temperature step in the "3He n_0 (pccSTP)" column.

Similarly to 3He, n_0^{4He} values decrease for every permeation step. Because the measurements of 4He are blank level, the evolution of n_0^{4He} is tracked in the "4He n0 (nccSTP)" column using Eq. (1.1) to account for long cooling times for some permeation steps. Because the permeation coefficient of 4He is not known for any temperature for Win-06-01b, the permeation coefficients for 3He at the relevant temperature is multiplied by 1.10 ± 0.19 (2σ) and the errors are all propagated in quadrature. Because of large blank corrections and uncertainty in the permeation coefficient, the errors for corrected 4He yields are very high. Regardless, diffusivities calculated from these 4He data form a sensible array on an Arrhenius plot. Many of these measurements are within error of the 3He array, but some less uncertain data are resolved to lower apparent diffusivity values, suggesting that there is some degree of helium loss recorded even up to 14% cumulate yield of 3He.

Naïve-Method Isothermal Permeability Dataset

The effective permeability coefficients used by the informed bracketing method below – and the isothermal-permeation-fit methodology, cooling-tube numerical model, and Arrhenius regression used to determine them – are described in full in the main text (§**??**). The complete fitted dataset underlying that regression is reported here: Table 1.1 gives the raw per-step data for all 10 isothermal sequences, and Table 1.2 gives the lambda/n_0 fit to each sequence (Fig. **??**) that together define the Arrhenius regression of Fig. **??**.

Table 1.1: Raw step-heat data ('DistilledNaiveLambdas' sheet) for the 10 isothermal 3He permeation sequences used to determine the ampoule's apparent permeability (Fig. **??**, Fig. **??**). Series are separated by horizontal rules; within a series, steps are listed in the order measured.

Series	Temp (°C)	Nominal Time (min)	Cooling Time (min)	^{3}He measured (pccSTP)	$\pm 2\sigma$
1	105.00	150	26	0.0190	1.1×10^{-4}
1	105.00	150	26	0.0111	7.4×10^{-5}
1	105.00	150	26	0.0080	5.3×10^{-5}
2	109.99	150	26	0.0229	1.4×10^{-4}
2	110.00	150	26	0.0115	7.3×10^{-5}
2	110.00	150	26	0.0075	5.2×10^{-5}
3	99.86	120	26	0.0106	6.1×10^{-5}
3	100.00	120	26	0.0074	4.0×10^{-5}
3	100.00	120	26	0.0049	2.5×10^{-5}
3	99.99	120	26	0.0043	2.1×10^{-5}
4	95.00	150	26	0.0109	2.8×10^{-4}
4	95.00	150	26	0.0081	2.0×10^{-4}
4	95.00	150	26	0.0062	1.4×10^{-4}
5	104.99	120	26	0.0215	8.9×10^{-5}
5	104.99	120	26	0.0144	6.0×10^{-5}
5	104.99	120	26	0.0104	4.3×10^{-5}
6	100.00	120	26	0.0126	5.6×10^{-5}
6	100.00	120	26	0.0094	3.5×10^{-5}
6	100.00	120	26	0.0074	2.0×10^{-5}
7	95.00	120	26	0.0072	3.0×10^{-5}

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Series	Temp (°C)	Nominal Time (min)	Cooling Time (min)	^{3}He measured (pccSTP)	$\pm 2\sigma$
7	95.00	120	26	0.0063	2.6×10^{-5}
7	95.00	120	26	0.0045	1.9×10^{-5}
8	90.00	150	26	0.0059	2.7×10^{-5}
8	89.90	150	26	0.0048	2.4×10^{-5}
8	90.00	150	26	0.0036	1.9×10^{-5}
9	90.00	180	26	0.0100	6.2×10^{-5}
9	89.92	180	26	0.0067	5.1×10^{-5}
9	90.00	180	26	0.0054	4.8×10^{-5}
10	99.99	150	26	0.0104	4.3×10^{-5}
10	99.98	150	26	0.0068	3.5×10^{-5}
10	100.00	150	26	0.0054	3.4×10^{-5}

Table 1.2: Fitted λ (permeation rate constant) and n_0 (residual helium at the start of the series) for each of the 10 isothermal sequences in Table 1.1, from the weighted non-linear least-squares fit of Eq. (**??**) (Chapter **??**) described in the main text (§**??**). n is the number of steps in the sequence. These are the same per-sequence fits plotted in Fig. **??** and used to build the Arrhenius regression in Fig. **??**.

Series	Temp (°C)	Duration (min)	n	λ (s^{-1})	$\pm 2\sigma$	n_0 (pccSTP)	$\pm 2\sigma$
1	105.00	150	3	4.9×10^{-5}	1.4×10^{-5}	0.0521	7.8×10^{-3}
2	109.99	150	3	6.3×10^{-5}	1.8×10^{-5}	0.0508	7.3×10^{-3}
3	99.86	120	4	4.3×10^{-5}	1.3×10^{-5}	0.0377	5.2×10^{-3}
4	95.00	150	3	3.1×10^{-5}	2.6×10^{-6}	0.0444	2.2×10^{-3}
5	104.99	120	3	5.0×10^{-5}	6.2×10^{-6}	0.0696	4.7×10^{-3}
6	100.00	120	3	3.7×10^{-5}	4.1×10^{-6}	0.0534	3.4×10^{-3}
7	95.00	120	3	3.3×10^{-5}	1.5×10^{-5}	0.0348	1.1×10^{-2}
8	90.00	150	3	2.7×10^{-5}	4.8×10^{-6}	0.0279	3.5×10^{-3}
9	90.00	180	3	3.0×10^{-5}	9.0×10^{-6}	0.0357	7.0×10^{-3}
10	99.99	150	3	3.8×10^{-5}	1.2×10^{-5}	0.0351	7.0×10^{-3}

Naïve-Method Step-Heat Dataset

The complete corrected step-heat dataset underlying the naïve bracketing method's apparent permeabilities (Fig. **??**) is reported in Tables 1.3 and 1.4, reporting the central value and 2σ uncertainty for each quantity, analogous to Tables **??** and **??** in the main text for the informed bracketing ("treatment as uncertainty") method. Table 1.5 separately reports the bulk-normalized ^{4}He/^{3}He ratios and cumulative fractional ^{3}He yield for each step. "–" denotes steps for which a quantity could not be determined (see §**??** in the main text).

Table 1.3: ^{3}He step-heat dataset for HT-Win-06-01b under the naïve bracketing method: measured and cumulative ^{3}He, the initial and residual ampoule content (n_0, n_t), the true step production (ω), and the apparent diffusivity $\ln(D/a^2)$, each with its 2σ uncertainty.

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	n_0 (pccSTP)	$\pm 2\sigma$	n_t (pccSTP)	$\pm 2\sigma$	ω (pccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
170	300	0.9630	0.0000	0.00×10^{0}	0.00×10^{0}	0.1968	2.22×10^{-2}	1.1597	2.63×10^{-2}	-21.910	0.045
180	463	0.5442	0.9630	7.26×10^{-2}	1.29×10^{-2}	0.0782	4.93×10^{-3}	0.5497	1.60×10^{-2}	-22.184	0.089
190	550	0.5292	1.6241	1.95×10^{-2}	2.54×10^{-3}	0.0679	8.20×10^{-3}	0.5777	1.30×10^{-2}	-21.976	0.085
200	500	0.5190	2.2112	3.28×10^{-2}	5.42×10^{-3}	0.0532	5.77×10^{-3}	0.5394	1.15×10^{-2}	-21.702	0.088
210	475	0.5770	2.7646	1.41×10^{-2}	2.11×10^{-3}	0.0552	5.71×10^{-3}	0.6182	1.17×10^{-2}	-21.311	0.082
220	450	0.6333	3.3799	1.10×10^{-2}	1.80×10^{-3}	0.0469	4.58×10^{-3}	0.6692	1.24×10^{-2}	-20.991	0.080
230	425	0.7899	4.0551	5.87×10^{-3}	1.86×10^{-3}	0.0639	3.48×10^{-3}	0.8479	1.25×10^{-2}	-20.514	0.066
225	908	0.7513	4.8885	1.95×10^{-2}	2.45×10^{-3}	0.0420	4.94×10^{-3}	0.7738	1.42×10^{-2}	-21.200	0.121
215	450	0.1415	5.6842	1.31×10^{-2}	1.85×10^{-3}	0.0278	3.54×10^{-3}	0.1563	6.48×10^{-3}	-22.015	0.316
205	520	0.0763	5.8529	7.18×10^{-3}	1.23×10^{-3}	0.0125	1.63×10^{-3}	0.0816	8.10×10^{-3}	-22.788	0.829
195	1106	0.0670	5.9491	2.12×10^{-3}	4.02×10^{-4}	0.0307	2.86×10^{-3}	0.0956	3.23×10^{-3}	-23.370	1.094
185	740	0.0227	6.0246	1.25×10^{-2}	1.64×10^{-3}	0.0090	1.06×10^{-3}	0.0192	2.00×10^{-3}	-24.567	2.822
175	920	0.0180	6.0610	3.68×10^{-3}	5.02×10^{-4}	0.0045	4.42×10^{-4}	0.0189	7.42×10^{-4}	-24.796	3.797
200	260	0.0224	6.0814	6.53×10^{-4}	1.17×10^{-4}	0.0095	1.23×10^{-3}	0.0313	1.65×10^{-3}	-23.022	1.495
210	520	0.0812	6.1081	7.48×10^{-3}	9.98×10^{-4}	0.0145	1.32×10^{-3}	0.0882	3.85×10^{-3}	-22.669	1.135
220	1167	0.5766	6.1914	2.82×10^{-3}	1.44×10^{-3}	0.0276	3.24×10^{-3}	0.6014	2.45×10^{-2}	-21.504	0.184
230	370	0.3577	6.7796	8.68×10^{-3}	3.96×10^{-3}	0.0422	5.13×10^{-3}	0.3912	1.48×10^{-2}	-20.712	0.146
240	320	0.6767	7.1563	1.72×10^{-2}	2.42×10^{-3}	0.0826	9.94×10^{-3}	0.7420	1.57×10^{-2}	-19.807	0.099
245	250	0.5549	7.8582	2.13×10^{-2}	8.87×10^{-3}	0.0516	5.65×10^{-3}	0.5852	1.46×10^{-2}	-19.710	0.125
250	210	0.6934	8.4743	1.08×10^{-2}	4.52×10^{-3}	0.0718	8.15×10^{-3}	0.7545	1.41×10^{-2}	-19.200	0.102
245	270	0.4659	9.2086	2.48×10^{-2}	3.48×10^{-3}	0.0498	5.84×10^{-3}	0.4909	1.01×10^{-2}	-19.810	0.197
240	320	0.2857	9.7206	2.07×10^{-2}	2.81×10^{-3}	0.0339	4.08×10^{-3}	0.2989	6.58×10^{-3}	-20.433	0.312
235	370	0.2199	10.0357	1.66×10^{-2}	2.20×10^{-3}	0.0273	3.29×10^{-3}	0.2307	5.23×10^{-3}	-20.810	0.401
230	420	0.1400	10.2736	1.32×10^{-2}	1.76×10^{-3}	0.0255	2.53×10^{-3}	0.1523	4.00×10^{-3}	-21.332	0.603
225	652	0.1494	10.4279	1.26×10^{-2}	1.62×10^{-3}	0.0131	1.47×10^{-3}	0.1499	3.42×10^{-3}	-21.772	0.747
230	500	0.1550	10.5908	7.34×10^{-4}	1.67×10^{-3}	0.0128	1.32×10^{-3}	0.1671	3.37×10^{-3}	-21.383	0.461
235	450	0.1951	10.7582	2.57×10^{-3}	4.18×10^{-4}	0.0334	1.85×10^{-3}	0.2259	3.29×10^{-3}	-20.956	0.364
240	400	0.2425	10.9638	6.90×10^{-3}	1.45×10^{-3}	0.0251	3.16×10^{-3}	0.2607	1.23×10^{-2}	-20.672	0.318
245	350	0.3150	11.2328	5.79×10^{-3}	1.05×10^{-3}	0.0395	4.64×10^{-3}	0.3487	9.39×10^{-3}	-20.219	0.240
250	300	0.3716	11.5689	1.65×10^{-2}	2.24×10^{-3}	0.0467	5.24×10^{-3}	0.4018	9.62×10^{-3}	-19.888	0.209
250	300	0.3640	11.9627	2.43×10^{-2}	4.37×10^{-3}	0.0339	3.82×10^{-3}	0.3735	7.99×10^{-3}	-19.927	0.242
250	300	0.3452	12.3490	1.13×10^{-2}	1.60×10^{-3}	0.0395	4.48×10^{-3}	0.3733	7.91×10^{-3}	-19.895	0.244
250	10250	8.4829	12.7167	1.91×10^{-2}	2.53×10^{-3}	0.1165	7.38×10^{-3}	8.5804	1.28×10^{-1}	-19.958	0.189

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(continued)

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	n_0 (pccSTP)	$\pm 2\sigma$	n_t (pccSTP)	$\pm 2\sigma$	ω (pccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
240	785	0.2261	21.2187	2.90×10^{-9}	6.85×10^{-9}	0.0263	2.92×10^{-3}	0.2525	2.21×10^{-2}	-20.660	0.580
Total			146.1483								

Table 1.4: ^{4}He step-heat dataset for HT-Win-06-01b under the naïve bracketing method, structured as in Table 1.3.

Temp (°C)	Time (min)	^{4}He meas. (nccSTP)	Cum. ^{4}He (nccSTP)	n_0 (nccSTP)	$\pm 2\sigma$	n_t (nccSTP)	$\pm 2\sigma$	ω (nccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
170	300	0.0192	0.0000	0.00×10^{0}	0.00×10^{0}	3.10×10^{-3}	0.0014	0.0169	0.0057	-30.366	0.674
180	463	0.0305	0.0192	1.14×10^{-3}	5.30×10^{-4}	3.99×10^{-3}	0.0011	0.0281	0.0055	-28.998	0.446
190	550	0.0558	0.0660	1.22×10^{-3}	3.80×10^{-4}	7.42×10^{-3}	0.0016	0.0586	0.0039	-27.556	0.233
200	500	0.1024	0.1375	3.77×10^{-3}	8.85×10^{-4}	1.12×10^{-2}	0.0023	0.1066	0.0044	-26.114	0.134
210	475	0.1460	0.2500	3.32×10^{-3}	8.39×10^{-4}	1.48×10^{-2}	0.0030	0.1517	0.0071	-25.110	0.112
220	450	0.2091	0.4055	3.40×10^{-3}	9.21×10^{-4}	1.64×10^{-2}	0.0033	0.2142	0.0095	-24.217	0.098
230	425	0.3243	0.6319	2.79×10^{-3}	8.21×10^{-4}	2.83×10^{-2}	0.0051	0.3430	0.0097	-23.222	0.073
225	908	0.3483	0.9799	1.39×10^{-2}	2.76×10^{-3}	1.46×10^{-2}	0.0034	0.3517	0.0076	-23.575	0.114
215	450	0.0716	1.3487	7.32×10^{-3}	1.73×10^{-3}	5.64×10^{-3}	0.0025	0.0747	0.0075	-24.245	0.318
205	520	0.0385	1.4359	3.01×10^{-3}	8.65×10^{-4}	8.98×10^{-3}	0.0083	0.0338	0.0120	-25.142	1.022
195	1106	0.0336	1.4912	9.47×10^{-4}	4.68×10^{-4}	8.68×10^{-4}	0.0026	0.0258	0.0180	-26.143	2.313
185	740	0.0079	1.5467	4.94×10^{-3}	4.56×10^{-3}	3.39×10^{-3}	0.0011	-0.0021	0.0080	–	–
175	920	0.0152	1.6020	3.82×10^{-4}	1.16×10^{-3}	4.93×10^{-3}	0.0011	0.0152	0.0034	-26.474	3.286
200	260	0.0107	1.6351	5.77×10^{-4}	2.20×10^{-4}	1.07×10^{-2}	0.0322	0.0149	0.0011	-25.224	2.214
210	520	0.2166	1.6547	3.95×10^{-3}	8.87×10^{-4}	1.20×10^{-2}	0.0055	0.0612	0.1666	-24.476	3.011
220	1167	0.3187	1.8714	2.95×10^{-3}	8.90×10^{-3}	2.52×10^{-2}	0.0059	0.2365	0.0940	-23.836	1.736
230	370	0.2113	2.6766	4.31×10^{-3}	2.06×10^{-3}	4.90×10^{-2}	0.0136	0.2146	0.0199	-22.653	1.000
240	320	0.4286	3.1616	1.11×10^{-2}	2.82×10^{-3}	3.93×10^{-2}	0.0082	0.4018	0.0683	-21.726	0.680
245	250	0.3988	3.6430	1.43×10^{-2}	4.47×10^{-3}	4.89×10^{-2}	0.0102	0.4084	0.0195	-21.290	0.660
250	210	0.4440	4.2359	1.13×10^{-2}	2.88×10^{-3}	3.58×10^{-2}	0.0075	0.4663	0.0204	-20.824	0.594
245	270	0.3123	4.7263	1.85×10^{-2}	4.43×10^{-3}	2.45×10^{-2}	0.0052	0.3210	0.0132	-21.326	1.050
240	320	0.1927	5.0846	1.61×10^{-2}	3.71×10^{-3}	2.02×10^{-2}	0.0043	0.1955	0.0090	-21.918	1.654
235	370	0.1517	5.3031	1.27×10^{-2}	2.87×10^{-3}	1.90×10^{-2}	0.0039	0.1548	0.0071	-22.249	2.064
230	420	0.0991	5.4714	1.04×10^{-2}	2.35×10^{-3}	8.87×10^{-3}	0.0020	0.1031	0.0067	-22.749	3.071
225	652	0.0995	5.5835	9.79×10^{-3}	2.15×10^{-3}	9.15×10^{-3}	0.0019	0.0909	0.0087	-23.291	4.243
230	500	0.1051	5.6969	3.22×10^{-3}	8.17×10^{-4}	2.42×10^{-2}	0.0047	0.1063	0.0056	-22.845	2.498
235	450	0.1369	5.8251	2.11×10^{-3}	5.73×10^{-4}	1.93×10^{-2}	0.0044	0.1502	0.0106	-22.363	1.886
240	400	0.1781	5.9762	4.79×10^{-3}	1.27×10^{-3}	3.09×10^{-2}	0.0065	0.1835	0.0115	-22.007	1.539
245	350	0.2305	6.1807	5.05×10^{-3}	1.47×10^{-3}	3.77×10^{-2}	0.0078	0.2493	0.0110	-21.519	1.128
250	300	0.2777	6.4385	1.39×10^{-2}	3.23×10^{-3}	2.66×10^{-2}	0.0055	0.2965	0.0110	-21.135	0.942
250	300	0.2654	6.7374	2.41×10^{-2}	5.21×10^{-3}	3.00×10^{-2}	0.0064	0.2619	0.0106	-21.204	1.142
250	300	0.2489	7.0231	9.79×10^{-3}	2.34×10^{-3}	1.02×10^{-1}	0.0186	0.2585	0.0150	-21.168	1.159
250	10250	6.8042	7.2901	1.54×10^{-2}	3.62×10^{-3}	2.15×10^{-2}	0.0054	6.8476	0.0767	-20.869	0.644
240	785	0.1853	14.1156	1.08×10^{-8}	3.61×10^{-8}	1.85×10^{-1}	0.0215	0.0054	0.0192	-21.540	1.209
Total			145.8145								

Table 1.5: Cumulative ^{3}He fractional yield and bulk-normalized ^{4}He/^{3}He ratios for HT-Win-06-01b under the naïve bracketing method. $(^4\text{He}/^3\text{He})_\omega$ is the ratio of the step's permeation-corrected ^{4}He and ^{3}He production (ω); F ^{3}He is the cumulative fractional ^{3}He yield; $(^4\text{He}/^3\text{He})_\omega$ is this step's ratio normalized to the bulk (whole-experiment) ^{4}He/^{3}He ratio. These data have much smaller uncertainties than those produced when permeation lag is treated as uncertainty.

Temp (°C)	Time (min)	$(^4\text{He}/^3\text{He})_\omega$	$\pm 2\sigma$	F ^{3}He	$\frac{(^4/^3\text{He})}{(^4/^3\text{He})_{\text{bulk}}}$	$\pm 2\sigma$
170	300	14.5470	4.91×10^{0}	0.0079	0.0146	0.0049
180	463	51.0439	1.02×10^{1}	0.0117	0.0512	0.0102
190	550	101.4852	7.20×10^{0}	0.0156	0.1017	0.0073
200	500	197.7263	9.21×10^{0}	0.0193	0.1982	0.0095
210	475	245.3349	1.24×10^{1}	0.0236	0.2459	0.0127
220	450	320.0824	1.54×10^{1}	0.0281	0.3208	0.0158
230	425	404.5353	1.29×10^{1}	0.0340	0.4055	0.0136
225	908	454.4434	1.29×10^{1}	0.0392	0.4555	0.0138
215	450	477.9049	5.18×10^{1}	0.0403	0.4790	0.0522
205	520	413.8402	1.53×10^{2}	0.0409	0.4148	0.1532
195	1106	269.9648	1.88×10^{2}	0.0415	0.2706	0.1887
185	740	–	–	0.0417	–	–
175	920	807.5143	1.84×10^{2}	0.0418	0.8094	0.1844
200	260	474.9353	4.38×10^{1}	0.0420	0.4760	0.0442
210	520	693.4629	1.89×10^{3}	0.0426	0.6951	1.8938
220	1167	393.2223	1.57×10^{2}	0.0467	0.3941	0.1575
230	370	548.6002	5.50×10^{1}	0.0494	0.5499	0.0554
240	320	541.4759	9.28×10^{1}	0.0545	0.5427	0.0932
245	250	697.8180	3.76×10^{1}	0.0585	0.6994	0.0385
250	210	618.0428	2.94×10^{1}	0.0636	0.6195	0.0302
245	270	653.8778	3.01×10^{1}	0.0670	0.6554	0.0310
240	320	654.1454	3.33×10^{1}	0.0690	0.6556	0.0341
235	370	671.0610	3.44×10^{1}	0.0706	0.6726	0.0352
230	420	677.0110	4.77×10^{1}	0.0717	0.6786	0.0484
225	652	606.4642	5.99×10^{1}	0.0727	0.6079	0.0604
230	500	636.2191	3.58×10^{1}	0.0738	0.6377	0.0365
235	450	664.7204	4.79×10^{1}	0.0754	0.6662	0.0486
240	400	704.0598	5.54×10^{1}	0.0772	0.7057	0.0560
245	350	714.8865	3.70×10^{1}	0.0795	0.7165	0.0379
250	300	737.8848	3.27×10^{1}	0.0823	0.7396	0.0337
250	300	701.0048	3.21×10^{1}	0.0849	0.7026	0.0330
250	300	692.4920	4.29×10^{1}	0.0874	0.6941	0.0436
250	10250	798.0564	1.49×10^{1}	0.1461	0.7999	0.0172
240	785	749.8748	1.01×10^{2}	0.1478	0.7516	0.1011
Total		997.716				

The finite-difference numerical model used to evaluate the contribution of the 26-minute cooling period to these apparent permeabilities – and to confirm that the ampoule's markedly elevated permeability is not an artifact of that cooling history – is described in full in the main text (§**??**).

Diffusivities

Apparent values of D/a^2 for 3He and 4He were calculated from times and corrected cumulative yields using the equations of **fechtigkalbitzer1966**. Uncertainties in the corrected values are propagated into the cumulative yield, and all errors are treated as uncorrelated for simplicity. Error bars plotted in the Arrhenius plots on this tab are likely exaggerated. Derivatives necessary to calculate uncertainty in the diffusivities using quadrature are tabulated to the right of the sheet. The Stage 1 reference-line linear regression (Fig. **??**), its statistics, and the treatment of total-yield uncertainty in the regression are described in the main text (§**??**); the same regression underlies both the main text's treatment-as-uncertainty results and the naive-bracketing dataset reported here.

Totals

The main text (§**??**) already summarizes this final furnace-fusion step and its blank/re-extract corrections; the additional spreadsheet-level detail below (volume-split calibration, standard-averaging convention) is retained here for completeness. For the final step, the glass ampoule was broken and the remainder of the helium was completely degassed in a resistance furnace at 1100°C. The gas from this final step was split within the vacuum line on calibrated volumes so that purified helium from less than 1% of the evolved gas was inlet into a Thermo Fisher Helix SFT mass spectrometer. A gas standard with a known amount of 3He and 4He was measured twice preceding the final fusion measurement of the hydrothermal Win-06-01b, as was a split furnace hot blank. The gas standards were used to calculate sensitivities, which were taken to be the mean of the two measurements, with 1σ uncertainty approximated by their standard deviation. After the split measurement of the Win-06-01b ampoule, the furnace was reheated to 1100°C to ensure complete extraction of the helium in the sample. The final step was then taken to be the measured value minus the hot blank, plus the re-extract, all divided by their respective volume split fractions and multiplied by the calculated sensitivity. Uncertainties were propagated in quadrature.